

AGENTS.md

SynChain Multi-Agent Architecture

Overview

SynChain uses specialized AI agents that collaborate through a centralized orchestrator.

Each agent owns a specific business responsibility.

Agent 1: Demand Forecast Agent

Goal

Predict future product demand.

Inputs

- Product Data
- Current Demand
- Historical Demand
- Market Signals
- News Signals

Outputs

- Demand Forecast
 - Growth Rate
 - Confidence Score
-

Agent 2: Inventory Agent

Goal

Optimize inventory levels.

Inputs

- Current Inventory
- Forecasted Demand

Outputs

- Recommended Inventory
 - Safety Stock
 - Overstock Risk
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Agent 3: Logistics Agent

Goal

Optimize movement of goods.

Inputs

- Warehouse Data
- Product Data
- Inventory Data

Outputs

- Warehouse Selection
 - Distribution Plan
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Agent 4: Risk Agent

Goal

Identify operational risks.

Inputs

- Supplier Data
- Inventory Data
- Logistics Data

Outputs

- Risk Score
 - Risk Category
 - Risk Explanation
-

Agent 5: News Intelligence Agent

Goal

Monitor external events.

Sources

- Industry News
- Supply Chain News
- Economic News

Outputs

- News Signals
 - Event Impact Score
-

Agent 6: Market Intelligence Agent

Goal

Analyze market conditions.

Outputs

- Market Trends
 - Demand Indicators
 - Opportunity Signals
-

Agent 7: Supplier Intelligence Agent

Goal

Evaluate supplier reliability.

Outputs

- Supplier Score
 - Delay Probability
 - Reliability Rating
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Agent 8: Scenario Simulation Agent

Goal

Simulate future possibilities.

Scenarios

- Demand Surge
- Supplier Failure
- Transportation Delay
- Warehouse Shutdown

Outputs

- Predicted Outcomes
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Agent 9: Recommendation Agent

Goal

Generate final decisions.

Outputs

- Strategic Recommendations
 - Prioritized Actions
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Agent 10: Explanation Agent

Goal

Explain recommendations.

Outputs

- Human-readable reasoning
 - Supporting evidence
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Agent 11: Orchestrator Agent

Goal

Coordinate all agents.

Responsibilities

- Workflow control
 - State management
 - Agent sequencing
 - Result aggregation
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Agent Execution Order

News Agent ↓ Market Agent ↓ Supplier Agent ↓ Demand Agent ↓ Inventory Agent ↓ Logistics Agent ↓ Risk Agent ↓ Scenario Agent ↓ Recommendation Agent ↓ Explanation Agent